

Test Cases related for Test plan

Student Record System

1. Purpose

This Test Plan defines the approach, scope and test cases used to verify that the Student Record System meets the functional and non-functional requirements defined in the Software Requirements Specification (SRS).

It supports the Requirements Traceability Matrix and Validation Specification already defined for this project.

2. Scope

Testing covers authentication and role-based access, student record management (add, view, search, update, delete), academic record management (grades), attendance management, search/filter operations, input validation, persistent storage, and the non-functional qualities defined for the system (usability, performance, security/access control).

3. Test Levels

- Unit Test (UT): Tests one function/class/module in isolation.
- Functional Test (FT): Checks whether a complete function satisfies its requirement.
- System Test (ST): Verifies end-to-end behaviour across integrated modules.
- Acceptance Test (AT): Confirms the system meets user/stakeholder expectations.

The test cases below are written at the Functional Test level unless noted, and map to the Test Case IDs already referenced in the Requirements Traceability Matrix (TC-FR-xx / TC-NFR-xx).

4. Test Case Summary

Test Case ID	Req. ID	Test Description / Steps	Expected Result	Type	Status
TC-FR-01	FR-01	Login with valid credentials for Administrator, Faculty and Student roles.	User is authenticated and directed to the role-based main menu.	Functional	Planned
TC-FR-02	FR-02, AF-01/AF-02	Login with an incorrect username or password, then with an empty username or password field.	System rejects the login, displays an error message (or a message requesting the required credentials) and allows the actor to re-enter credentials.	Functional	Planned

Test Case ID	Req. ID	Test Description / Steps	Expected Result	Type	Status
TC-FR-03	FR-03, FR-04	Log in as each role and verify the main menu only shows permitted operations.	Each actor sees only the functions permitted for their role (role-based menu).	Functional	Planned
TC-FR-06	FR-06, AF-01	Administrator adds a new student with a Student ID that already exists.	System rejects the new record and displays a duplicate Student ID message.	Functional	Planned
TC-FR-09	FR-09, AF-01	Search for a student using a Student ID or name that does not exist.	System displays a message indicating no matching student record was found.	Functional	Planned
TC-FR-10	FR-10, AF-01/AF-02	Update a student record with a non-existent Student ID, then with invalid data.	System displays a not-found message, or rejects invalid changes and requests valid input.	Functional	Planned
TC-FR-11	FR-11, AF-02	Administrator deletes a student record and cancels at the confirmation prompt.	System retains the student record and returns to the previous menu.	Functional	Planned
TC-FR-12	FR-12, AF-02	Faculty records marks/grades outside the permitted range.	System rejects the value and requests valid marks.	Functional	Planned
TC-FR-15	FR-15, AF-01	Faculty records an attendance value outside the permitted range.	System rejects the value and requests valid attendance data.	Functional	Planned
TC-NFR-02	NFR-02	Perform add, search, update and view operations on 1,000 student records.	Each operation completes within 2 seconds.	Non-Functional	Planned
TC-NFR-03	NFR-03	Attempt to add a duplicate Student ID and to save invalid record changes, then review the stored records.	Duplicate IDs are prevented and stored records remain consistent.	Non-Functional	Planned

Test Case ID	Req. ID	Test Description / Steps	Expected Result	Type	Status
TC-NFR-08	NFR-08	Attempt to modify student records using a role without modification rights (e.g. Student).	System prevents the unauthorized modification and denies the operation.	Non-Functional	Planned

5. Pass / Fail Criteria

A test case passes when the system produces the Expected Result exactly as specified for the given input and role.

Any deviation, unhandled error, or incorrect access-control behaviour is recorded as a Fail, logged with reproduction steps, and linked back to its Req. ID for correction before re-test.

6. Status Tracking

Each test case's Status will be updated from 'Planned' to 'Pass' or 'Fail' after execution, consistent with the Status column used in the Requirements Traceability Matrix.